

Corvinna Curtis

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Assignment: CS 470 Final Reflection

YouTube Presentation: <https://youtu.be/QVAGAwILanM>

Experiences and Strengths

Taking CS 470 was one of the most hands-on and valuable experiences I've had so far in my computer science program. I didn't just learn how to build a web application, I learned how to migrate it to the cloud and make it scalable, reliable, and modern. Working through each stage of development helped me gain confidence in my ability to handle both the frontend and backend sides of a project, as well as understand how everything connects through cloud services.

I developed and strengthened several technical skills throughout this course. I learned how to containerize a full-stack MEAN application using Docker and Docker Compose, which gave me a clear understanding of how to package and deploy code consistently. Then I moved into the cloud environment, where I learned how to work with AWS S3, Lambda, API Gateway, and DynamoDB. I now have a better understanding of how to use serverless services to build applications that scale automatically without constant management.

This course helped me grow as a software developer by forcing me to think beyond just writing code. I had to consider how my application would run in the real world, how users would access it, how data would be handled securely, and how costs would change as it scaled. I'd say one of my biggest strengths as a developer is adaptability. I can pick up new technologies quickly, troubleshoot issues methodically, and see the bigger picture behind what I'm building.

Because of this experience, I feel prepared to take on roles such as ServiceNow Developer, Cloud Engineer, or Full Stack Developer. I'm comfortable managing both front-end design and back-end logic, and I understand the value of automation and scalability in production environments. I can confidently explain my technical decisions to both technical and non-technical audiences, which is a key skill in my career field.

Planning for Growth

Planning growth of my web application is where everything I've learned about cloud computing really comes together. One of the biggest advantages of using AWS is that I can

scale my application without having to rebuild it from scratch. Using serverless functions and microservices makes management and scaling much more efficient.

If my application gained more users in the future, I would handle scale by increasing concurrency limits for my Lambda functions and setting up monitoring with AWS CloudWatch to track performance and errors. For error handling, I'd use retries and dead-letter queues, so failed requests don't get lost.

When it comes to predicting cost, AWS serverless is generally more predictable for smaller or variable workloads, since you only pay for what you use. However, if my application had steady, heavy traffic, containerized solutions like ECS or EKS might offer better long-term cost control. Serverless shines for flexibility, while containers can sometimes be cheaper for constant high usage.

For expansion, I would look at a mix of both approaches depending on the need. Serverless functions are great for quick, event-driven tasks, while containerized services are ideal for long-running processes. The main deciding factors would be cost, complexity, and control.

Elasticity and the pay-for-service model would continue to play a big role in decision making. Elasticity ensures that my app can handle sudden spikes in traffic automatically, and the pay-for-service model allows me to scale at my own pace without wasting resources. Together, they create an environment that supports sustainable growth while keeping costs manageable, two things that are essential for modern web development and for my career goals moving forward.

Final Thoughts

This course brought everything together for me, from coding to deployment to real-world problem solving. I'm leaving CS 470 not only with a finished project but also with the confidence that I can build, scale, and maintain applications in the cloud. It's exactly the kind of foundation I need to continue growing in my career as a developer.